

1. Batch

16 MATH prompts x
topic $c(x)$ = subject

topic

Macro gate (CB)

topic budget from past
steps' positive advantage

g_i^{mac}

2. Sample

policy π_θ : $G=8$ answers
per prompt (+1 greedy)

answers

Micro gate

SELF: skip
solved prompts
or MEG: rare modes up

g_i^{mic}

3. Reward

$r_i \in \{0, 1\}$
(Math-Verify)

4. Advantage

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r + \epsilon}$$

$\hat{A}_i$

5. Update $\tilde{A}_i = g_i^{\text{mac}} g_i^{\text{mic}} \hat{A}_i \rightarrow$ PPO-clip loss, update the LoRA weights